Wenpin Hou, Ph.D.

Date of Preparation: October 5, 2025

PERSONAL DATA

Name:	Wenpin Hou
Contact Information:	Room 650, 722 West 168th Street, New York, NY 10032, USA

ACADEMIC APPOINTMENTS / WORK EXPERIENCE

Assistant Professor *(tenure-track)*, Columbia University, New York, NY, USA 7/2022-Present
Department of Biostatistics, Mailman School of Public Health
Affiliated Member, Data Science Institute, Columbia University, New York, NY, USA 1/2024-Present
Center for Foundations of Data Science
Center for Health Analytics

EDUCATION

Ph.D., Mathematics, The University of Hong Kong, Hong Kong, China 9/2013-8/2017
Mentor: Dr. Wai-Ki Ching
Thesis: Mathematical modelling and optimization in biological networks and data
Degree Awarded Date: 11/25/2017
B.Sc., Information and Computational Science, Sun Yat-sen University, Guangzhou, China 9/2009-6/2013
Advisor: Dr. Tianshou Zhou
Degree Awarded Date: 6/19/2013

TRAINING

Postdoctoral Fellow
Johns Hopkins University, Department of Biostatistics, Baltimore, MD, USA 8/2019-6/2022
Mentors: Drs. Stephanie C. Hicks and Hongkai Ji
K99 Phase Mentors: Drs. Hongkai Ji, Stephanie C. Hicks, and Andrew P. Feinberg
K99 Phase Advisors: Drs. Xiaobin Wang, Kasper Hansen, and Gregory Hager
Johns Hopkins University, Department of Computer Science, Baltimore, MD, USA 8/2017-8/2019
Mentors: Drs. Suchi Saria and Aravinda Chakravarti

Visiting Researcher
NYU Langone Health, Center for Human Genetics and Genomics, New York, NY, USA 12/2018-12/2018
MD Anderson Cancer Center, Department of Bioinformatics & Computational Biology, Houston, TX, USA 7/2017-8/2017
Kyoto University, Bioinformatics Center, Kyoto, Japan 5/2016-8/2016
Kyoto University, Bioinformatics Center, Kyoto, Japan 5/2015-8/2015
Soka University, Bioinformatics Lab, Tokyo, Japan 6/2014-8/2014

HONORS AND AWARDS

Maximizing Investigators Research Award (MIRA) for Early Stage Investigators, NIH/NIGMS 2023
Emerging Leaders in Computational Oncology Award, Memorial Sloan Kettering Cancer Center 2021
NIH Pathway to Independence Award (K99/R00), NIH/NHGRI 2021
2021 Women in Statistics and Data Science Conference Award, American Statistical Association 2021
First-place Winning Poster at the 13th Annual Symposium and Poster Session on Genomics and Bioinformatics 2019
University Postgraduate Fellowships, Philip K H Wong Foundation, The University of Hong Kong 2013-2017
Postgraduate Scholarship, The University of Hong Kong 2013-2017
Excellent Teaching Award, Department of Mathematics, The University of Hong Kong 2016
Doris Chen Postgraduate Travel Grants 2016
Xu Chongqing Award for Academic Excellence (*two students awarded among Philosophy, Math, Physics, and Chemistry Schools*) 2012
Academic Award for Excellent Undergraduates (*received annually for three years, including one First-Class distinction*) 2010-2012

ADMINISTRATIVE LEADERSHIP AND ACADEMIC SERVICE

Institutional Service

Co-found the Statistical Genetics and Genomics (SGG) Working Group 9/2022

• Form a group of 35 sign-up members and in charge of the mailing list

• Organize bi-weekly SGG seminars and invite six external speakers from UPenn, Duke, UMich, UNC Chapel Hill, NCI, Boston University, etc. from 9/2023 to 12/2024

Departmental Committees

• Member, Research Advisory Committee 9/2024-Present
• Member, PhD Admission Committee 10/2023-Present
• Member, Application Qualifying Exams Committee 10/2023-Present
• Member, Curriculum Committee 10/2023-Present

Summer Programs

• Mentor of three fellows, Columbia Summer Research Institute (CSRI) 6-8/2025
• Mentor of two Master's students, SIBDS program 6/2024

Miscellaneous

• Seminar speaker, PhD student recruitment visit day 2/2025
• Seminar speaker, PhD student recruitment visit day 2/2024
• Panelist, Doctoral Job Search: Academia workshop, Columbia Mailman School of Public Health, Virtual 10/2023
• Speaker and poster judge, Columbia Biostatistics Annual Research Symposium (CBARS) 9/2023

External Service

• Panelist, ASA Panel Discussion on K99/R00 Grants in Statistical Genetics and Genomics 9/2025

PROFESSIONAL ORGANIZATIONS AND SOCIETIES

Memberships

Institute of Mathematical Statistics (IMS) 2023-2024
International Chinese Statistical Association (ICSA) 2023-2024
American Statistical Association (ASA) 2021-2022, 2023-2026
Eastern North American Region (ENAR) 2019-2020, 2025
International Society for Computational Biology (ISCB) 2018-2019
Society for Industrial and Applied Mathematics (SIAM) 2014-2018

NIH Grant Reviewer

Special emphasis panel (SEP) COBRE Phase 1 ZRG1 MGG-M (40) P 7/2025
NCI Special Emphasis Panel ZCA1 TCRB-9: U01/U24 Review 4/2025
Maximizing Investigators' Research Award - F Study Section (MRAF) 2/2025
Genomics, Computational Biology and Technology Study Section (GCAT) 10/2024

Journal Reviewer

Bioinformatics (5), Biometrics (2), Briefings in Bioinformatics, Communications Biology, Genomics, Proteomics & Bioinformatics, Genome Biology (3), IEEE Transactions on Neural Networks and Learning Systems, Journal of the American Statistical Association (JASA), NAR Genomics and Bioinformatics, Nature Communications (17), Nature Genetics, Nature Methods (4), Nature Portfolio, Neurocomputing, PLoS Computational Biology (4), Precision Nutrition

Conference Award Committees

• The 12th International Chinese Statistical Association Conference (ICSA2023)
• International Conference on Health Policy Statistics (ICHPS 2023)

GRANT SUPPORT ***The * indicates the grants cannot be accepted currently because of over-funded status***

Active Research Funding

DSI Seed Fund, Columbia University, MPI (Wei, Noble, and Hou), 0 CM 7/2025-6/2026

Title: TRANSFORM-AD: A Pilot Transformer-based AI Platform for Personalized Alzheimer's Disease Progression Forecasting and Intervention Learning

Major Goals: The project aims to develop a prototype of an AI platform, TRANSFORM-AD, which uses advanced transformer models and comprehensive nationwide Alzheimer's data to forecast disease trajectories at the individual level and develop utility tools to uncover key mechanistic insights and guide personalized care. This pilot will evaluate the platform's potential to provide robust, trustworthy AI-driven solutions that enhance Alzheimer's research, improve treatment strategies, and support precision healthcare.

Direct Cost: $125,000

R35GM150887 (R01-equivalent), NIH/NIGMS, **PI**, 5.508 CM 9/2023-8/2028
Title: Methods for inferring and analyzing gene regulatory networks using single-cell multiomics and spatial genomics data
Major Goals: To develop computational methods to infer dynamic and spatial gene regulatory networks from single-cell multiomics data and design control strategies to identify minimal driver genes for manipulating cell states.
Total Cost: $ 2,031,164

K99/R00HG011468, NIH/NHGRI, **PI**, 5.292 CM 3/2021-6/2026
Title: Computational methods for inferring single-cell DNA methylation and its spatial landscape
Major Goals: To develop computational methods to predict and reconstruct single-cell and spatial DNA methylation landscapes, enabling accurate characterization of epigenomic variability and therapeutic targets in disease.
Total Cost: $1,030,861 (actually used $914,853)

1R01MH133561-01*, NIH/NIMH, co-I(PI: Boldrini, M.), 0.6 CM 9/2023-6/2028
Title: Human brain multi-omics to decipher major depression pathophysiology
Major Goals: To create a comprehensive molecular and spatial atlas of the human hippocampus in major depressive disorder by integrating proteomics, single-cell multiomics, and spatial transcriptomics, in order to uncover pathogenic mechanisms and identify novel therapeutic targets.
Total Cost: $3,863,494

Pending Support
MIND Prize, The Pershing Square Foundation, **PI**, 2.4 CM 3/2026-2/2029
Title: NeuroSpatial-3D — A Generative Foundation Model for 3D Spatial Genomics of the Aging and Alzheimer's Disease Brain
Total Cost: $750,000
Status: Submitted on 9/7/2025

Michelson Prize, Michelson Foundation, **PI**, 0 CM 1/2026-12/2027
Title: ImmuAgent: An Autonomous AI Agent for Multimodal Immune Profiling and Vaccine Discovery
Total Cost: $150,000
Status: Submitted on 6/11/2025

R01, NIH/DHHS, **PI**, 4.8 CM 4/2026-11/2030
Title: Foundation Models for Multimodal Single-Cell Analysis and Perturbation Response Prediction
Total Cost: $2,828,693
Status: Submitted on 6/3/2025

R01, NIH/DHHS, co-I (PI: Zhang, X.), 0.6 CM 12/2025-11/2030
Title: Wnt-FGF Signaling Crosstalk Regulates Eye Development
Total Cost: $3,465,840
Status: Competitive renewal, submitted on 2/28/2025

Past Support
R01AG076949 Supplement*, NIH, co-I(PI: Boldrini, M.), 3 CM 12/2023-11/2024
Title: Harmonization and joint analysis of human brain single-cell datasets from neurotypical aging controls and alzheimer's disease patients
Major Goals: To harmonize and integrate multiomic and spatial datasets from aging and Alzheimer's disease brains with large-scale public resources to build a comprehensive reference atlas of human brain cell types, uncover gene regulatory networks driving neuropsychiatric symptoms, and identify targets for new therapeutic strategies. Total Cost: $366,174

EDUCATIONAL CONTRIBUTIONS

Direct Teaching
Instructor, Department of Biostatistics, Columbia University
- P8131 Biostatistical Methods II Spring 2024, 2025
  - Sole instructor for this mandatory course, providing three hours of lectures per week

  - Curriculum includes Generalized Linear Models, Longitudinal Data Analysis, and Survival Analysis
  - Master's and PhD level students were enrolled per year
  - 113 enrolled students, Spring 2024
  - 104 enrolled students, Spring 2025

Guest Lecturer
- Graduate Student Research Seminar, Department of Biostatistics Fall 2024
- Graduate Student Research Seminar, Department of Biostatistics Fall 2023

Teaching Assistant and Lab Instructor, Department of Mathematics, The University of Hong Kong 9/2013-2/2017
Certificate: Certificate of Teaching and Learning in Higher Education
Responsibility: give supplement lectures, programming lessons, and exercise class twice per week; design and grade assignments; provide instruction to individual or small groups of students to improve academic performance; advise students about their academic and career development
- MATH 1011 University Mathematics (**1** semester)
- MATH 2601&3601 Numerical Analysis (**2** semesters)
- MATH 1211&2211 Multivariable Calculus (**3** semesters)

Advising and Mentorship (The ‡ indicates that I am the primary mentor or practicum advisor.)
Postdoctoral Research Scientists, Department of Biostatistics, Columbia University
- Xu Liao‡ 2/2025-Present
- Qi Liu, (co-mentored by Dr. Ying Wei) 10/2024-Present
- Won Eui Hong‡, (co-mentored by Dr. Ying Wei) 2/2024-Present

NIH Awards Postdocs
- Wending Li, Postdoc at Department of Environmental Health Sciences, Columbia University, mentored through K99/R00 (submitted). Role: Advisor 10/2025-Present
- Alicia Arredondo Eve, Postdoc at the University of Illinois Urbana-Champaign, mentored through NIH/NIGMS Training Grant - Career MODE (R25GM143298). Role: Advisor 9/2025-6/2026

Fellows, Mentored through Columbia Summer Research Institute, New York, NY, USA
- Jacqueline J. Tao, MD, Chief Fellow, Columbia University Irving Medical Center 7/2025-8/2025
- Gaston Jean-Louis, Jr., MD, Chief Fellow, NewYork-Presbyterian Hospital /Columbia University Irving Medical Center 7/2025-8/2025
- Naama Elefant, MD,PhD, Research associate, Center for Precision Medicine and Genomics (CPMG) at Columbia University 7/2025-8/2025

PhD Students, Department of Biostatistics, Columbia University, New York, NY, USA
- Chhiring Y. Lama (Y2024) 9/2024-Present
- Elly Kipkogei (Y2023, Provost diversity fellow), mentor through SPRIS II course 9/2023-1/2024

Master's Students, Department of Biostatistics, Columbia University, New York, NY, USA
- Qingyu Xiong‡ MA Student (Y2024-2026, Advanced Machine Learning track) in Statistics, focusing on statistical modeling and large language models for fine-grained, emotion-aware sentiment analysis. 6/2025-Present
- Xun Sun‡ (Y2024, Data Science Track) 6/2025-Present
- Zitao Zhang, (Y2024, Theory and Methods Track) 9/2024-Present
  - Accepted by GEMS program at Memorial Sloan Kettering Cancer Center
- Xinyi Shang‡, (Y2023, Theory and Methods Track) 5/2024-Present
  - Chair's Award for research excellence; Contributed to two papers where I am the senior author
- Lehan Zou‡ (Y2023,Theory and Methods Track) 9/2023-Present
- Aiying Huang‡ (Y2023, Public Health Data Science Track) 9/2023-Present
- Yifei Zhao‡, (Y2022, Theory and Methods Track), current computer engineer at DISH 10/2023-10/2024
  - Contributed to one preprint where I am the senior author
- Jingyi Yao‡ (Y2022,Theory and Methods Track), current PhD student at Boston University 3/2023-5/2024
- Tianchuan Gao‡ (Y2021, Theory and Methods Track), current PhD student at Indiana University 9/2022-5/2023
- Wenhan Bao (Y2021, Theory and Methods Track), current PhD student at Florida University 8/2022-6/2023

Master's Students, Department of statistics, Columbia University, New York, NY, USA

- Qingyu Xiong (Y2024-2026, Advanced Machine Learning track) 6/2025-Present

Undergraduate Students
- Matthew Eichner, Pomona College, mentor through SIBDS program 6/2024-Present
- Alexandra Duta, University of Illinois Urbana-Champaign, mentor through SIBDS program 6/2024-7/2024

Professional Education
- Panelist, Harvard Medical School - Chinese Students and Scholars Association (HMS-CSSA) Career Development Series, Virtual 8/2022
- Panelistm Chan Zuckerberg Initiative (CZI) Faculty Search Bootcamp: Recent Jobseeker Panel, Virtual 8/2022

PUBLICATIONS ***Symbols indicate my independent work (white numbers), mentees (__), and senior authors (*).***

Peer-Reviewed Research Publications:

Generative Pretrained Transformer (GPT) Models in Biomedical Applications

1. **Shang, X.**, **Liao, X.**, Ji, Z., and **Hou, W.***, 2025. Benchmarking large language models for genomic knowledge with GeneTuring. Briefings in Bioinformatics, 26(5), p.bbaf492. Software (GPT-s App): SeqSnap.
2. **Hou, W.***, **Liu, Q.**, Ma, H., Qu, Yi. and Ji, Z.*, 2024. Assessing large multimodal models for one-shot learning and interpretability in biomedical image classification. Advanced Intelligent Systems, April 6, 2025.
3. **Hou, W.***, and Ji, Z.*, 2024. Comparing large language models and human programmers for generating programming code. Advanced Science, 30 December 2024.
4. **Hou, W.***, and Ji, Z.*, 2024. Assessing GPT-4 for cell type annotation in single-cell RNA-seq analysis. Nature Methods, 2024 March 25. **[IF: 48].** Software: GPTCelltype.
   Note 1: This highly cited paper received enough citations to place it in the **top 1%** of the academic field of Biology & Biochemistry based on a highly cited threshold for the field and publication year. **Note 2**: Featured in 12 news outlets, such as Columbia News Spotlight, Columbia MSPH News, Science Daily, The Medical News, Health Tech World, etc. **Note 3**: Reviewed in Nature Methods "Embedding AI in biology" and "Toward learning a foundational representation of cells and genes".

Method development for Single-cell Genomics and Epigenomics

5. **Hou, W.**, Ji, Z., Chen, Z., Wherry, E.J., Hicks, S.C. and Ji, H., 2023. A statistical framework for differential pseudotime analysis with multiple single-cell RNA-seq samples. Nature Communications 14, 7286. Software: Lamian.
6. Wang, Y., Wang, W., Liu, D., **Hou, W.**, Zhou, T. and Ji, Z., 2023. GeneSegNet: A deep learning framework for cell segmentation by integrating gene expression and imaging. Genome Biology 24, 235. Software: GeneSegNet.
7. **Hou, W.** and Ji, Z., 2022. Palo: Spatially-aware color palette optimization for single-cell and spatial data. Bioinformatics, June 01, 2022. Software: Palo. PMID: 35642896.
8. **Hou, W.** and Ji, Z., 2022. Single-cell unbiased visualization with SCUBI. Cell Reports Methods, 100135, 2022. Software: SCUBI. PMID: 35224531. PMCID: PMC8871596.
   Note 1: Discussed in Tumour antigen-induced T cell exhaustion – the archenemy of immune-hot malignancies in Nature Reviews Clinical Oncology 18, 749–750 (2021).
   Note 2: I contributed to the pseudotime analysis and functional analysis (Figure 4g-h, Extended data Figure 12b-f).
9. **Hou, W.**, Ji, Z., Ji, H. and Hicks, S.C., 2020. A systematic evaluation of single-cell RNA-sequencing imputation methods. Genome Biology 21, 218. PMID: 32854757. PMCID: PMC7450705.
10. Ji, Z., Zhou, W., **Hou, W.** and Ji, H., 2020. SCATE: single-cell ATAC-seq signal extraction and enhancement. Genome Biology, 21,161. PMID: 32620137. PMCID: PMC7333383. Software: SCATE and SCATEData.

Method Applications in Omics Study

11. Lo, E.K., Idrizi, A., Tryggvadottir, R., Zhou, W., **Hou, W.**, Ji, H., Cahan, P. and Feinberg, A.P. 2024. DNA methylation memory of pancreatic acinar-ductal metaplasia transition state altering Kras-downstream PI3K and Rho GTPase signaling in the absence of Kras mutation. Genome Medicine, 17(1), p.32. **Note: I contributed to the WGBS and Visium data generation through my NIH K99/R00HG011468 grant.**
12. Jackson, C., Cherry, C., Bom, S., Dykema, A., Wang, R., Thompson, E., Zhang, M., Li, R., Ji, Z., **Hou, W.**, Li, R., Zhang, H. and Choi, J., Rodriguez, F., Weingart, J., Yegnasubramanian, S., Lim, M., Bettegowda, C., Powell, J., Eliesseff, J., Ji, H., and Pardoll, D. 2025. Distinct myeloid derived suppressor cell populations promote tumor aggression in glioblastoma. Science, 387, 6713. **Note: I contributed to the single-cell trajectory inference and dynamic genes detection (Figure 2d-e).**
13. Montagne, J.M. Mitchell, J.T., Tandurella, J.A., Christenson, E.S., Danilova, L.V., Deshpande, A., Melanie L., Sidiropoulos, D.N., Davis-Marcisak, E.,Bergman, D.R., Zhu, Q., Wang, H., Kagohara, L.T., Engle, L.L., Green, B.F., Favorov, A.V., Ho, W.J., Lim, S.J., Zhang, R., Li, P., Gai, J., Mo, G., Mitchell, S., Wang, R., Vaghasia, A., **Hou, W.**, Xu, Y.,

Zimmerman, J.W., Elisseeff, J.H., Yegnasubramanian, S., Anders, R.A., Jaffee, E.M., Zheng, L. and Fertig, E.J. 2024. CD137 agonism enhances anti-PD1 induced activation of clonally expanded CD8+ T cells in a neoadjuvant pancreatic cancer clinical trial. iScience. **Note: I contributed to the TCR-seq data alignment and enrichment analysis.**

14. Dykema, A.G., Zhang, J., Cheung, L.S., Connor, S., Zhang, B., Zeng, Z., Cherry, C.M., Li, T., Caushi, J.X., Nishimoto, M., Munoz, A.J., Ji, Z., **Hou, W.**, Zhan, W., Singh, D., Zhang, T., Rashid, R., Mitchell-Flack, M., Bom, S., Tam, A., Ionta, N., Aye, T.H.K., Wang, Y., Sawosik, C.A., Tirado, L.E., Tomasovic, L.M., Spangler, J.B., Anagnostou, W., Yang, S., Spicer, J., Rayes, R., Taube, J., Brahmer, J.R., Forde, P.M., Yegnasubramanian, S., Ji, H., Pardoll, M., and Smith K.N., 2023. Lung tumor–infiltrating Treg have divergent transcriptional profiles and function linked to checkpoint blockade response. Science Immunology, 8(87). PMID: 37713507. PMCID: PMC10629528.
15. Caushi, J.X., Zhang, J., Ji, Z., Vaghasia, A., Zhang, B., Hsiue, E., Mog, B., **Hou, W.**, Justesen, S., Blosser, R., Tam, A., Anagnostou, V., Cottrell, T.R., Guo, H., Chan, H., Singh, D., Thapa, S., Dykema, A., Choudhury, C., Aparicio, L., Cheung, L., Lanis, M., Belcaid, Z., Asmar, M.E., Illei, P., Brock, M., Ha, J., Bush, E., Park, B., Bott, M., Naidoo, J., Marrone, K.A., Reuss, J.E., Velculescu, V.E., Chaft, J.E., Kinzler, K.W., Zhou, S., Vogelstein, B., Taube, J.M., Merghoub, T., Brahmer, J.R., Hellmann, M.D., Forde, P.M., Yegnasubramanian, S., Ji, H., Pardoll, D.M., Smith, K.N., 2021. Transcriptional programs of neoantigen-specific TIL in anti-PD-1-treated lung cancers. Nature 596, 126–132. PMCID: PMC8338555.

Controllability in Boolean Networks

16. **Hou, W.**, Ruan, P., Ching, W.K. and Akutsu, T., 2019. On the number of driver nodes for controlling a Boolean network when the targets are restricted to attractors. Journal of Theoretical Biology, 463, pp.1-11. PMID: 30543810.
17. **Hou, W.**, Tamura, T., Ching, W.K. and Akutsu, T., 2016. Finding and analyzing the minimum set of driver nodes in control of Boolean networks. Advances in Complex Systems, 19(03), p.1650006. PMID: 30679639. PMCID: PMC6345816.

Glycosylation Networks Construction

18. **Hou, W.***, Qiu, Y., Hashimoto, N., Ching, W.K. and Aoki-Kinoshita, K.F., 2016. A systematic framework to derive *N*-glycan biosynthesis process and the automated construction of glycosylation networks. BMC Bioinformatics, 17(7), p.240. PMID: 27454116. PMCID: PMC4965717.

Statistical Methods for Public Health Research

19. Xu, R., Hong, X., Ladd-Acosta, C., Buckley, J.P., Choi, G., Wang, G., **Hou, W.**, Wang, X., Liang, L. and Ji, H., 2023. Contrasting Association of Maternal Plasma Biomarkers of Smoking and 1-Carbon Micronutrients with Offspring DNA Methylation: Evidence of Aryl Hydrocarbon Receptor Repressor Gene–Smoking–Folate Interaction. The Journal of Nutrition, 153(8), 2339-2351.
20. Huang, W., Igusa, T., Wang, G., Buckley, J.P., Hong, X., Bind, E., Steffens, A., Mukherjee, J., Haltmeier, D., Ji, Y., Xu, R., **Hou, W.**, Fan, Z., and Wang, X., 2022. In-utero co-exposure to toxic metals and micronutrients on childhood risk of overweight or obesity: new insight on micronutrients counteracting toxic metals. International Journal of Obesity, 46, 1435–1445.
21. **Hou, W.***, Zhang, M., Ji, Y., Hong, X., Wang, G., Xu, R., Liang, L., Saria, S. and Ji, H., 2022. A prospective birth cohort study of maternal prenatal ciga-rette smoking assessed by self-report and biomarkers on childhood risk of overweight or obesity. Precision Nutrition, 1(3), e00017.
22. Xu, R., Hong, X., Zhang, B., Huang, W., **Hou, W.**, Wang, G., Wang, X., Igusa, T., Liang, L. and Ji, H., 2021. DNA methylation mediates the effect of maternal smoking on offspring birthweight: a birth cohort study of multi-ethnic US mother–newborn pairs. Clinical Epigenetics, 13(1), pp.1-13. PMID: 33663600. PMCID: PMC7931602.
23. Ji, Y., Azuine, R.E., Zhang, Y., **Hou, W.**, Hong, X., Wang, G., Riley, A., Pearson, C., Zuckerman, B. and Wang, X., 2019. Association of cord plasma biomarkers of in utero acetaminophen exposure with risk of attention-deficit/hyperactivity disorder and autism spectrum disorder in childhood. JAMA psychiatry, 77(2), pp.180-189.. PMID: 31664451. PMCID: PMC6822099. (**Featured in NIH news, Reuters health, MedPage Today, meaww, LinksMedicus, TechnologyNetworks**)

Machine Learning for Biomedical Data

24. Jiang, H., Qiu, Y., **Hou, W.**, Cheng, X., Yim, M.Y. and Ching, W.K., 2018. Drug side-effect profiles prediction: from empirical to structural risk minimization. IEEE/ACM transactions on computational biology and bioinformatics, 17(2), pp.402-410. PMID: 29994681.
25. Jiang, H., Ching, W.K., Cheung, W.S., **Hou, W.** and Yin, H., 2017. Hadamard kernel SVM with applications for breast cancer outcome predictions. BMC Systems Biology, 11(7), p.138. PMID: 29322919. PMCID: PMC5763304.
26. Jiang, H., Ching, W.K. and **Hou, W.**, 2016. On orthogonal feature extraction model with applications in medical prognosis. Applied Mathematical Modelling, 40(19-20), pp.8766-8776.

27. Qiu, Y., Cheng, X., **Hou, W.** and Ching, W.K., On classification of biological data using outlier detection, The 12th International Symposium on Operations Research and its Applications (ISORA), pp.144-150, Luoyang, China, 2015.
28. Cheng, X., Ching, W.K., **Hou, W.** and Kinoshita, K., A multiple linear regression model for structure of *N*-linked oligosaccharides, The 12th International Symposium on Operations Research and its Applications (ISORA), pp.151-157, Luoyang, China, 2015.
29. Cheng, X., Qiu, Y., **Hou, W.** and Ching, W.K., A semi-tensor product approach for probabilistic Boolean networks, IEEE Proceedings of the 8th International Conference on Systems Biology (ISB), pp.88-93, Qingdao, China, 2014.

Thesis

30. **Hou, W.** (2017) Mathematical modelling and optimization in biological networks and data. Ph.D. Thesis. The University of Hong Kong.

Meetings/Invited Oral and Poster Presentations:

Posters

31. CSHL Systems Immunology 2021, Virtual. "A computational framework for differential pseudotime analysis across conditions with multiple single-cell RNA-seq samples reveals T cell immune dynamics associated with COVID-19 disease severity." 4/2021
32. 13th Annual Symposium and Poster Session on Genomics and Bioinformatics, Johns Hopkins University, Baltimore, MD, USA. "A systematic evaluation of single-cell RNA-seq imputation methods." 10/2019 (awarded as *first-place winning poster*)
33. 11th annual RECOMB/ISCB Conference on Regulatory & Systems Genomics with DREAM Challenges, New York University, New York, NY, USA. "Causal gene regulatory network construction using single-cell RNA-seq and single-cell ATAC-seq data." 12/2018
34. Machine Learning Summer School, University of Cadiz, Cadiz, Spain. "On orthogonal feature extraction model with applications in cancer prediction." 5/2016

PRESENTATIONS

Invited Seminar Talks

1. University of Pennsylvania, Department of Biostatistics, Epidemiology and Informatics. Pennsylvania, USA. 4/29/2025
2. Vanderbilt University, Department of Biostatistics. Nashville, USA. 4/8-9/2025.
3. Boston University, Statistical Genetics & Genomics at Department of Biostatistics. *"Expanding the horizons of genomics and biomedical analysis: GPT-4's role in automated cell type annotation and beyond"*. Virtual. 12/6/2024
4. Memorial Sloan-Kettering Cancer Center, Department of Epidemiology and Biostatistics. *"Expanding the horizons of genomics and biomedical analysis: GPT-4's role in automated cell type annotation and beyond"*. New York, USA. 6/26/2024
5. Statistical Genetics and Genomics Working Group at Columbia Biostatistics. *"Statistical methods for deciphering cellular trajectories in single-cell and spatial transcriptomics data"*. 9/20/2023.
6. Johns Hopkins University, Department of Biostatistics, Statistical Genetics Working Group. Baltimore, MD, USA. 12/9/2021
7. Johns Hopkins University, Department of Biostatistics. Baltimore, MD, USA. 3/2019
8. MD Anderson Cancer Center, Department of Bioinformatics and Computational Biology. Houston, TX, USA. 7/2017
9. Kyoto University, Bioinformatics Center. Kyoto, Japan. 3/2016
10. Kyoto University, Bioinformatics Center. Kyoto, Japan. 6/2015
11. Soka University, Department of Bioinformatics. Tokyo, Japan. 8/2014

Invited Panel Discussion

1. Adapting to AI in Healthcare: an in-Person Workshop, Columbia University Irving Medical Center. *"Open AI in Basic & Translational Ressearch — Opportunities in Health & Diseases"*. 4/29/2024
2. Data Science for Public Health Summit, Columbia University. *"AI transforming science: genomics, imaging, and prediction"*. 1/31/2025

Invited Job Talks

1. Duke University, Department of Biostatistics and Bioinformatics. 2/18/2025

2. The University of North Carolina at Chapel Hill, Department of Biostatistics. 2/6/2025
3. University of Michigan, Department of Computational Medicine and Bioinformatics. 3/8/2022
4. Duke University, Department of Biostatistics and Bioinformatics. 2/22/2022
5. Columbia University, Department of Biostatistics. 2/16/2022
6. Mount Sinai, Department of Genetics and Genomic Sciences. 2/10/2022
7. New York University, Department of Biostatistics. 2/7/2022
8. UNC Chapel Hill, Department of Mathematics. Chapel Hill, NC, USA. 2/3/2022
9. Weill Cornell Medicine, Department of Population Health Sciences. 2/1/2022
10. New York University, NYU Langone Institute for Systems Genetics. 1/18/2022
11. University of Pittsburgh, Department of Biological Sciences. 12/2021
12. University of Southern California, Division of Biostatistics at Keck School of Medicine. 10/2021
13. University of Southern California, Center for Craniofacial Molecular Biology. 9/2021

Conference Talks

Invited:

1. Invited session, ENAR 2026 Spring Meeting, in Indianapolis, IN. *"Advances in AI-powered methods for genomic data science*. 3/15-18/2026
2. Invited session, ENAR 2025 Spring Meeting, Orleans, LA. *"Leveraging AI for Epigenomic Data Inference and Analysis"*. 3/23-26/2025
3. International Conference on Intelligent Biology and Medicine (ICIBM 2024), Houston Medical Center. 10/10-12/2024
4. Annual Workshop in Computational Biology, University of Chicago and Toyota Technological Institute at Chicago. 9/10/2024
5. Annual Synthetic Regeneration Symposium, Columbia University. New York, USA. *"Differential pseudotime analysis and spatio-temporal analysis in single-cell data"*. 7/14/2023
6. 2023 ICSA Applied Statistics Symposium. University of Michigan Ann Arbor, USA. *"Statistical methods development in single-cell genomics and multi-omics data"*. 6/14/2023
7. Columbia Stem Cell Initiative (CSCI) Retreat. Edith Macy Retreat Center, Briarcliff Manor, USA. *"Computational methods for spatial and temporal trajectory analysis in single-cell data"*. 5/4/2023
8. Emerging Leaders Symposium in Computational Oncology, Memorial Sloan Kettering Cancer Center, Virtual. *"Computational methods for analyzing gene expression and regulation dynamics at single-cell and spatial resolution"*. 10/14/2021

Contributed:

1. ENAR 2024 Spring Meeting, Baltimore, USA. *"Decomposing spatial heterogeneity of cell trajectories with Paella"*. 3/10/2024
2. 2021 Joint Statistical Meetings (JSM), Virtual. *"Lamian: a statistical framework for differential pseudotime analysis in multiple single-cell RNA-seq samples"*. 8/2021
3. ENAR 2020 Spring Meeting, Virtual. *"A systematic evaluation of single-cell RNA-seq imputation methods"*. 3/2020
4. Biotechnology and Bioinformatics Symposium (BIOT) 2015, Provo, UT, USA. *"A systematic framework to derive N-glycan biosynthesis process and the automated construction of glycosylation networks"*. 12/2015

Institutional seminar talks

Columbia Inter-disciplinary Collaboration Club, “Computational methods for single-cell and spatial genomic analysis", ICRC 816, 11/6/2024